

Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

2-AB, 2-aminobenzamide; ADCC, antibody-dependent cell-mediated cytotoxicity; ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete,

consistent, enduring and available; AMV, analytical method validation; ANOVA, analysis of variance; CDC, complement-dependent cytotoxicity; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, process capability index; CQA, critical quality attribute; DoE, design of experiments; DO, dissolved oxygen; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; iVCC, initial viable cell concentration; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; pCO₂, dissolved carbon dioxide partial pressure; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; UPLC, ultra-performance liquid chromatography; WC-CPP, well-controlled critical process parameter.

1 Purpose and scope

This plan describes the process characterization studies that will be performed on the A-Mab production bioreactor before the drug substance process is qualified at the receiving site (Novacyte Biologics — Grafton, WI (Commercial DS)). The bioreactor is Step 3 of the drug substance process and is operated as a fed-batch cell culture at 15,000 L. It is the step at which the glycosylation, charge variant and aggregate attributes of A-Mab are formed, and because the glycan attributes in particular are not modified by the platform purification train, the design space established here carries most of the product quality argument for the upstream process.

The studies will be executed on a qualified scale-down model at the sending site (Novacyte Biologics — Cambridge, MA (Development)). Their results, the design space that follows from them and the classification of every parameter will be reported in the paired report (PCR-003). Nothing in this plan states an outcome. It states what will be run, how the data will be analysed, and the criteria against which the analysis will be judged.

The work follows the enhanced, quality by design development approach, and it forms part of the Stage 1 process design record for the drug substance (U.S. Food and Drug Administration 2011; International Council for Harmonisation 2009, 2012). The scope of the study was set by the pre-characterization risk assessment (RA-001), and the campaign-level framework it sits inside, which covers the common scale-down approach, the common statistical approach and the acceptance framework, is given in the master plan (PCMP-001).

The plan covers the 9 process parameters of the production bioreactor listed in Table 6.1, of which 5 will be studied in a multivariate design and 4 will be assessed one at a time, in accordance with the study assignment made in RA-001 and not re-derived here. It also covers the 7 quality attributes that the drug substance register assigns to this step (Table 4.1), of which 5 are measured as design responses.

Four things are outside the scope of this plan. Vial thaw, seed expansion and the N-1 seed bioreactor are not characterized, because prior knowledge and the risk assessment place no product quality risk on them (Section 4.3). Harvest and clarification are covered by the harvest plan (PCP-004). The clearance of host cell protein (HCP) and residual DNA by the purification train is covered by PCP-005, PCP-007 and PCP-008. Confirmation of the design space at commercial scale belongs to Stage 2, and it follows the process performance qualification strategy set out in the transfer plan (PTP-001).

2 Objectives

The study has five objectives, and each one produces a defined section of PCR-003.

1. Quantify the relationship between the multivariate process parameters and the quality attributes this step forms, over ranges wider than the normal operating ranges (NORs).
2. Define the design space for the well-controlled culture parameters, which is the multivariate region within which every attribute in Table 4.1 is predicted to meet its acceptance criterion.
3. Establish a proven acceptable range (PAR) for each parameter against each attribute it governs, and confirm that the normal operating range sits inside it.
4. Provide the demonstrated effects and the control capability needed to classify each parameter under the classification procedure (SOP-4001).
5. Justify the parameter ranges that will be carried into Stage 2, and state this step's contribution to the control strategy for the drug substance.

The first two objectives depend on a design of experiments, the third on the fitted response-surface model, and the last two on both of those together with prior knowledge. If the scale-down model fails its qualification (Section 5.1), none of the five can be met, which is why qualification is the first activity in the sequence given in Section 11.

3 Related documents

Table 3.1 lists the documents that scope this plan and the documents it feeds. The three parent documents (PTP-001, RA-001 and PCMP-001) fix the transfer scope, the study scope and the campaign framework respectively.

Table 3.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the controlled procedures that govern the work, from scale-down model qualification (SOP-1001) through the operation of the step itself (SOP-2003) to parameter classification (SOP-4001). The validated analytical methods are listed separately in Table 5.1, together with the response each one supports.

Table 3.2: Standard operating procedures applicable to this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP

4 Prior knowledge and quality risk basis

Prior knowledge and RA-001 together fix the scope of this study. The platform is established, so the studies described below confirm and bound known behaviour instead of discovering it. What prior knowledge does not establish is how the culture parameters interact over ranges wider than routine control.

4.1 Unit-operation description and prior knowledge

The production bioreactor is a fed-batch cell culture in a stirred tank, operated with controlled pH, temperature, dissolved oxygen (DO) and nutrient feeding, and harvested at a fixed culture age. A-Mab is a humanized IgG1 antibody whose primary mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC), so the glycan attributes formed in the culture are linked to efficacy (CMC Biotech Working Group 2009). The glycosylation and charge variant distributions are established inside the cell and in the culture fluid, and neither is substantially modified by the platform purification train, which means that the values delivered at harvest are close to the values that reach drug substance.

Novacyte has operated this platform for three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and through the A-Mab clinical and engineering campaigns. That experience establishes the mechanisms this study expects to see, and it is the reason the study can be designed around 5 parameters instead of the full parameter list. The expectations are set out below so that the results reported in PCR-003 either confirm them or contradict them.

Culture pH acts on the intracellular and Golgi pH, and therefore on the activity of the glycosyltransferases and glycosidases that finish the glycan. It also shifts lactate metabolism and the integral of viable cell concentration, so it is expected to move several responses at once. Culture temperature governs growth rate and specific productivity, and a lower temperature slows growth, extends the production phase and raises the integral of viable cell concentration, which is the route by which it reaches the glycan profile. Dissolved carbon dioxide accumulates with scale and with the sparging strategy, and elevated $p\text{CO}_2$ lowers intracellular pH and reduces both the integral of viable cell concentration and the titre (CMC Biotech Working Group 2009). Osmolality rises through feeding and base addition, and is expected to act on specific productivity and on glycan processing.

Culture duration is expected to carry the largest effect on afucosylation and galactosylation, and on the impurity load delivered to harvest. A longer culture means that a larger fraction of the product is made by an ageing, less viable population, with

released glycosidases in the fluid and reduced nucleotide sugar availability inside the cell, so both the glycan profile and the aggregate burden are expected to move with harvest day. The same ageing raises the host cell protein and DNA load delivered to harvest (CMC Biotech Working Group 2009). High mannose behaves differently from the other two glycan attributes, because it reflects incomplete processing of the precursor and not the addition of a terminal residue, and it is therefore expected to respond to whatever changes the residence time of the glycoprotein in the Golgi (principally culture pH).

This step does not act alone on every attribute in its register. Aggregate formed here is polished principally at cation exchange (PCP-007), the host cell protein and DNA burden is cleared at Protein A, cation exchange and anion exchange (PCP-005, PCP-007 and PCP-008), and the low-pH hold contributes a small aggregate increase of its own (PCP-006). The bioreactor sets the input to that train, and the acceptance argument for the two process impurities is completed downstream, not here.

4.2 Quality attributes in scope

Table 4.1 lists the quality attributes that the drug substance register assigns to this step, with the acceptance criterion and the criticality of each.

Table 4.1: Quality attributes formed at the production bioreactor, with acceptance criteria and criticality. Tool #1 is the product of impact and uncertainty from the attribute assessment (CMC Biotech Working Group 2009).

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2-13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10-40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3-10 % of glycans	VH	80
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

Criticality is treated as a continuum and not as a binary split (International Council for Harmonisation 2023). Attributes at the high and very high levels are called critical, and they are the ones that drive the design of this study. High mannose is the only attribute at the very high level, and it carries the highest Tool #1 score in the table,

because a high impact score is compounded by genuine uncertainty about the level at which the attribute affects clearance. Afucosylation and galactosylation sit at the high level and are linked to potency through ADCC and complement-dependent cytotoxicity (CDC) respectively. Aggregates are also at the high level, and their acceptance limit is taken from prior X-Mab clinical experience.

The table also lists 3 attributes that are in scope for a reason other than their criticality. Acidic charge variants (deamidation) score at the very low level, but the bioreactor establishes the baseline charge variant distribution and the downstream steps change it only slightly, so the response is retained in the design and modelled with the others. Host cell protein and residual DNA are generated by the culture and are therefore assigned to this step, but neither is a design response of this study, because their acceptance is achieved by clearance in the purification train and their burden at harvest is characterized under PCP-004 and PCP-005.

All 5 responses are modelled here, because the culture sets the level of each one. The 3 glycan responses (afucosylation, galactosylation and high mannose) are not modified by the purification train, so their values at harvest are the values that reach drug substance. Acidic variants are changed only slightly downstream, and aggregate is polished principally at cation exchange (PCP-007), so for those two responses this step sets the input to the purification train and not the final level.

4.3 Risk-based prioritization of parameters

RA-001 assigned each parameter of this step to a study type by risk ranking and filtering, and this plan executes that assignment. Each parameter was ranked twice, once on its potential impact on a quality attribute (the main effect) and once on its potential to interact with other parameters, and the two rankings were combined into a study decision (CMC Biotech Working Group 2009). Where no data or rationale existed to support an assessment, the parameter was ranked at the highest level.

The outcome is 5 parameters carried into the multivariate design and 4 assessed one at a time. Culture pH, culture temperature, dissolved carbon dioxide, osmolality and culture duration all act on the glycan and charge variant chemistry through the mechanisms given in Section 4.1, and all 5 are expected to interact, which is what places them in a single factorial design instead of a series of one-factor studies. Dissolved oxygen and initial viable cell concentration (iVCC) act through growth and the integral of viable cell concentration and not through the glycan processing chemistry itself, so a one-at-a-time assessment bounds them adequately. Basal medium concentration and nutrient feed-1 volume are set by recipe and delivered by weight, and platform studies have found the effect of medium concentration on afucosylation to be statistically detectable but shallow across the full range (CMC Biotech Working Group 2009). Nutrient feed-1 volume acts through the same nutrient availability and is expected to behave in the same way.

The characterization ranges in Table 6.1 are deliberately wider than the normal operating ranges, and the expansion serves three purposes. It bounds the knowledge space, so that the severity of a parameter excursion can later be judged against data

instead of against judgement. It supports movement within the design space once the process is licensed, which is the practical value of a wide characterized region. It also gives the response-surface model enough leverage to resolve curvature, which a design run only over the routine control range cannot do, because the signal there is small against the analytical noise. The width of each range follows from the mechanism given in Section 4.1 and from the normal operating range it surrounds, and every parameter is studied beyond that range at both edges (Table 6.1). Platform experience bounds the upper edges in particular. A culture held at high pH and high temperature to a late harvest loses viability quickly at the end of the run, and that condition delivers the highest host cell protein and DNA burden to harvest (CMC Biotech Working Group 2009).

5 Materials and methods

5.1 Scale-down model and its qualification

Every study in this plan will be run on a bench-scale stirred tank bioreactor system that has been qualified as a model of the commercial vessel under SOP-1001. The conclusions of the study are applied directly to the commercial process, so qualification of the model is essential and is not a formality (U.S. Food and Drug Administration 2011).

The bench-scale and commercial vessels are stirred tanks with equivalent design characteristics (mixing, aeration and mass transfer) and equivalent process control capability (pH, dissolved oxygen, temperature and nutrient addition). For the qualification runs the scale-independent parameters are operated at the commercial target values, and these include pH, temperature, initial viable cell concentration, dissolved oxygen and culture duration. The scale-dependent parameters are treated differently. Agitation, gas flow rates, pressure, working volume and the resulting $p\text{CO}_2$ are established at small scale so that process performance matches the full-scale vessel (CMC Biotech Working Group 2009).

Qualification will compare replicate bench-scale runs at the target condition against the at-scale data set from the A-Mab engineering and clinical campaigns. The comparison covers the growth and productivity profile (peak viable cell density, viability at harvest and titre) and the 5 product quality responses. Each compared attribute is judged by an equivalence assessment against a pre-defined acceptance margin, so that a difference between scales is separated from run to run variability. The margins and the comparison itself are defined in the qualification protocol issued under SOP-1001 and are not restated here. Qualification will be complete and approved before any characterization run is executed.

The centre-point runs built into both designs serve a second purpose beyond the design itself. Their replicate spread is the pure error estimate against which lack of fit is tested (Section 5.5), and it is also a running check that the scale-down model has not drifted between the screening campaign and the response-surface campaign.

5.2 Cell line, media and culture operation

The culture will be inoculated from the A-Mab working cell bank and expanded through the platform seed train to the inoculation density given in Table 6.1. Basal medium and feeds are chemically defined and are prepared under SOP-2004. The production culture is operated fed-batch under SOP-2003, with feed additions on a

fixed schedule expressed as a percentage of working volume, pH controlled by carbon dioxide sparging and base addition, and dissolved oxygen controlled by the gas mix. The remaining multivariate factors are set indirectly. Dissolved carbon dioxide is imposed by the composition and flow of the sparge and overlay gas, and osmolality by the feed and base addition strategy. Both are verified against the daily metabolite panel described in Section 5.4.

Raw materials and inputs that are controlled to platform specification are not characterization factors. Medium and feed lots, gas supply, base solution and the working cell bank are identity and quality controlled on receipt and on use, and their control belongs to the routine control strategy and not to this study. Any lot change during execution will be recorded against the affected runs.

5.3 Analytical methods

Table 5.1 lists the validated methods that will generate the study data. Method performance, including accuracy, precision, specificity and range, is reported in the corresponding validation report and is not reproduced in PCR-003.

Table 5.1: Validated analytical methods supporting the study responses.

Reference	Title	Type
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Released N-glycan mapping by 2-AB HILIC-UPLC (SOP-3010, validated under AMV-3010) reports afucosylation, galactosylation and high mannose from a single separation, so those 3 responses share an assay and share its variance. Aggregates are measured by SEC-HPLC (SOP-3011, AMV-3011) and reported as the high molecular weight percentage. Acidic charge variants are measured by icIEF (SOP-3013, AMV-3013). Host cell protein by ELISA (SOP-3012, AMV-3012) and residual DNA by qPCR (SOP-3014, AMV-3014) will be run on the harvested culture fluid to document the burden delivered to the purification train, and, as stated in Section 4.2, neither is a design response here.

Assay variance matters to the interpretation of small effects. Where an estimated effect is of the same order as the reproducibility of its assay, PCR-003 will say so, and will attribute the variation to the method and not to the parameter. The centre-point replicate spread reported in PCR-003 is the basis for that judgement, and the method validation reports listed in Table 5.1 give the independent estimate against which it is checked.

5.4 Sampling plan

Each run will be sampled daily for viable cell density, viability and the routine metabolite panel (glucose, lactate, ammonium, osmolality and $p\text{CO}_2$), which supports the process performance narrative and confirms that each run held the factor settings it was assigned. The product quality panel is taken at harvest from clarified culture fluid, so that every response is measured on material in the same state and the comparison across runs is not confounded by the clarification step.

Both designs carry replicated centre points at the set-point condition. The screening design has 3 centre-point runs and the response-surface design has 4, which give 2 and 3 degrees of freedom for pure error respectively. Run order will be randomized within each campaign under SOP-1002, so that any drift in the analytical methods or in the seed expansion is not aliased with a factor. Retained samples will be stored for each run under the site retention policy.

5.5 Statistical methods

All analysis will be performed on coded factors. Each factor is coded so that its set-point is zero and the edges of its characterization range are minus one and plus one, which makes the estimated effects directly comparable across parameters measured in different units (pH units, degrees Celsius, mmHg, mOsm and days). The analysis will be performed under SOP-1002.

The screening data will be fitted by ordinary least squares to a model containing the 5 main effects and the 10 two-factor interactions. An effect is reported as twice the coded coefficient, which is the change in the response across the full range of the factor. Significance is judged at $\alpha = 0.05$. That model estimates 16 terms from 19 runs and leaves 3 residual degrees of freedom, so it is close to saturated. A near-saturated model identifies which factors are active, and it is not used to predict. The division is deliberate and is carried through the whole analysis. The screening design identifies the effects, and the response-surface model is the predictive model and the basis of the design space.

The response-surface data will be fitted to a full quadratic model in the 4 factors carried forward, containing the main effects, the two-factor interactions and the 4 pure quadratic terms. That model estimates 15 terms from 28 runs and leaves 13 residual degrees of freedom, which is enough to support prediction and to test the fit. Model adequacy will be assessed on the overall F test, on R^2 , on adjusted R^2 , and on predicted R^2 computed from the PRESS statistic. It will also be assessed by an ANOVA partition of the residual into lack of fit and pure error, using the centre-point replicates described in Section 5.4. A model whose lack of fit is significant against pure error will not be used to define the design space.

A parameter with no active term in either model is not discarded. It is reported as a null result, classified on that basis under SOP-4001, and retained in the knowledge space as evidence that the attribute is robust to it over the range screened. A parameter that

is inactive across the whole characterization range supports a stronger robustness statement than one shown to be inactive across its normal operating range alone.

Commercial-scale capability will be estimated by Monte-Carlo simulation of 2,000 batches at the normal operating condition. Each of the 4 factors of the response-surface model is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its normal operating range, and every draw is constrained to the characterization range. The fitted model then predicts each response for every simulated batch. The parameters the model does not contain are held at their set-points in the simulation, and their contribution is bounded separately by the univariate assessment (Section 6.4).

Capability will be reported as a Cpk for each response, with the margin between the predicted mean and the limit that binds. Afucosylation, galactosylation and high mannose are judged against two-sided acceptance criteria, so their Cpk is the smaller of the two one-sided indices, the one against the lower limit and the one against the upper limit. Acidic variants and aggregates are judged against an upper limit only (Section 8), so their Cpk is the one-sided index against that limit. The two proven acceptable range analyses are described separately in Section 8, because they answer a different question from capability.

6 Study design

The design answers two questions in sequence. A screening study resolves which of the 5 multivariate parameters are active on which response, and a response-surface study over the active subset produces the predictive model. The univariate work (Section 6.4) runs alongside and bounds the parameters that the risk assessment did not send to the factorial design.

6.1 Factors, ranges and study type

Table 6.1 gives the set-point, the normal operating range and the characterization range of every parameter of this step, with the study type assigned in RA-001.

Table 6.1: Production bioreactor parameters, ranges and study type. The range studied is the characterization range and is not a proven acceptable range.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
Dissolved CO2 (pCO2)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% sat	50	30–70	40–60	univariate
Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	X	1.2	0.8–1.6	1–1.4	univariate
Nutrient feed-1 volume	% WV	12	9–15	10–14	univariate

The two ranges in the table do different jobs, and PCR-003 will keep them apart. The normal operating range is where the commercial process will be held, and it sets the dispersion of the capability simulation (Section 5.5). The range studied is the

knowledge space edge, and it is the range over which the fitted models are valid. The design space lies between the two. It is the part of the characterized space in which every attribute is predicted to meet its criterion (Section 7), and the normal operating range of every parameter has to sit inside it.

Culture pH is held to 6.75–6.95 in routine operation and will be studied over 6.6–7.1. Culture temperature is held to 34.5–35.5 °C and studied over 34–36 °C. Dissolved carbon dioxide moves from a normal range of 70–130 mmHg to a studied range of 40–160 mmHg, because pCO₂ accumulates with scale and is not held by a control loop at commercial scale (it follows vessel height, sparge rate and cell density), so its excursions have to be bounded by data. Culture duration will be studied over 15–19 days around a set-point of 17 days, which brackets both an early and a late harvest decision.

Table 6.2 gives the letter code used for each factor in the design matrices and in the effect tables of PCR-003. The letters follow the screening order and are reused unchanged for the response-surface terms, so that a term such as AB means the same pair of factors in both analyses. Its last column states the anticipated response-surface subset described in Section 6.3, and that subset is subject to the screening result.

Table 6.2: Factor legend for the coded design matrices.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO ₂ (pCO ₂)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

6.2 Screening design

The screening study is a two-level fractional factorial in the 5 multivariate factors, augmented with 3 centre points, for 19 runs in total. The 16 factorial runs are a half fraction of the 32 run full factorial, chosen at resolution V so that every main effect and every two-factor interaction is estimable clear of the others. Two-factor interactions are aliased with three-factor interactions, which the platform history gives no reason to expect at this step (see Section 12 for how that assumption is managed). The planned matrix is given in Table 14.1.

The purpose of the screening study is to identify the parameters that have the largest effect on each quality attribute, and to identify which pairs of parameters interact. It is not intended to describe curvature and it will not be used to predict. Runs will be executed in a randomized order, and where an operational constraint requires the runs to be split into blocks, the block will be recorded and tested as a term in the model.

6.3 Response-surface design

The response-surface study is a face-centred central composite design in the subset of factors that screening identifies as active. It comprises 16 factorial runs, 8 axial runs at the faces of the cube and 4 centre points, for 28 runs in total. Placing the axial points on the faces keeps every run inside the characterization range. A rotatable design would put its axial runs outside the ranges given in Table 6.1, and the fitted model is declared only over the ranges characterized. The planned matrix is given in Table 14.2.

The design has been sized for 4 factors. Prior knowledge indicates that culture pH, culture temperature, culture duration and dissolved carbon dioxide are the factors most likely to be carried forward, and Table 14.2 is drawn for that subset. If the screening analysis identifies a different set of active factors, the design will be re-specified on the same principles before execution, and the change will be recorded against this plan.

This design adds the pure quadratic terms and therefore resolves curvature within the characterized ranges. It is the model that will be used to predict mean response levels, to draw the response surfaces, to derive the proven acceptable ranges (Section 8) and to define the design space reported in PCR-003.

6.4 Univariate assessment

Dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume will each be assessed one at a time. Each parameter will be run at the low and high edges of its characterization range with every other parameter at its set-point, and the results will be compared against the centre-point condition of the multivariate design. The same analytical panel applies, so a univariate result is directly comparable with a factorial result on the same response.

These 4 parameters were not placed in the factorial design because the risk assessment scored them below the multivariate threshold, and the reasons are mechanistic. Dissolved oxygen is held to 40–60 % sat by a control loop with ample authority, and it acts on the culture through growth and not through the glycan chemistry. Initial viable cell concentration is set at inoculation, and its effect on the culture is largely absorbed within the first days of a culture lasting 17 days. Basal medium concentration and nutrient feed-1 volume are delivered by weight against a recipe, and prior platform work has bounded their effect on the glycan profile as shallow (CMC Biotech Working Group 2009). A univariate range statement is the appropriate level of evidence for a parameter of that kind, and PCR-003 will report it as such rather than implying that it carries the weight of a multivariate result.

7 Acceptance and decision criteria

This section states the decisions that will be taken on the data before the data exist. Every criterion below has to be met for the study to support the design space and the parameter classifications that PCR-003 will report.

The scale-down model is qualified when the bench-scale runs at the target condition are shown to be equivalent to the at-scale data for the growth and productivity profile and for the 5 quality responses. Equivalence is assessed against a pre-defined acceptance margin on each compared attribute, and the margins are defined in the qualification protocol issued under SOP-1001 (Section 5.1). Characterization runs executed before that qualification is approved will not be used to define the design space.

A fitted model is adequate when the overall F test is significant at $\alpha = 0.05$, when the lack of fit is not significant against pure error, and when adjusted R^2 and predicted R^2 are consistent with each other. A large gap between adjusted and predicted R^2 indicates a model that fits the runs it was fitted to and predicts new conditions poorly, and such a model will be reduced to its significant terms before it is used. A model that fails on lack of fit will be reported with its diagnostics and will not be used to define the design space.

The design space will be declared over that part of the characterized space in which the fitted response-surface model predicts every attribute in Table 4.1 to lie within its acceptance criterion. The region will be reported in the planes of the factor pairs that carry the largest interactions, and the worst-case corner of the region, which is the combination of settings that comes closest to a limit, will be identified and stated. The normal operating range must lie inside the declared region for every parameter. If it does not, the finding will be reported as an edge of failure, and the normal operating range will be narrowed instead of the region being widened.

Capability will be reported as a Cpk per attribute with the margin to the binding limit, computed as described in Section 5.5. This plan sets no numerical minimum for Cpk. The acceptance framework for capability is given in PCMP-001 and applies across the campaign.

A parameter will be classified under SOP-4001 from two inputs, the demonstrated effect of the parameter on a quality attribute and the capability of the commercial equipment to hold it inside the design space (CMC Biotech Working Group 2009). A parameter with a demonstrated effect on a critical attribute and a high risk of falling outside the design space is a critical process parameter (CPP). One with a demonstrated effect but a low risk of falling outside the design space is a well-controlled critical process parameter (WC-CPP). A parameter that affects process performance without affecting a quality attribute is a key process parameter (KPP), and one with no demonstrated

effect on either is a general process parameter (GPP). The classification itself is an outcome, and it is reported in PCR-003.

An excursion from this plan during execution will be recorded as a deviation, investigated and impact assessed before the affected data are used. A run whose executed condition still falls inside the characterized ranges may be retained with a documented justification and analysed at its executed setting, because the fitted model is estimated from the conditions actually run. A run whose condition falls outside those ranges, or whose input material is not representative of the process, invalidates the affected design for the purpose of defining the design space, and that design will be re-executed in full. Both dispositions, and the reasoning behind whichever is applied, will be reported in the deviations section of PCR-003.

The study passes when the scale-down model is qualified, the models are adequate, the design space contains the normal operating range for every parameter, every attribute meets its acceptance criterion across that region, and every parameter carries a documented classification. The conclusions will be bounded to the ranges characterized, to the fitted models and to the qualified scale-down model, and they will not be extended to conditions outside the studied space.

8 Proven acceptable ranges (planned analysis)

The design space is a multivariate region, and a reader also needs the range of each parameter taken on its own. PCR-003 will therefore report a proven acceptable range for every combination of a governed attribute and a response-surface parameter, derived from the fitted model by the analysis described here.

The acceptance basis is the drug substance criterion recorded for each attribute in the quality attribute register (Table 4.1). Afucosylation, galactosylation and high mannose are judged against two-sided criteria, and acidic variants and aggregates against an upper limit only. The risk to the product from both of those attributes is an elevated level. The lower edge recorded for acidic variants in Table 4.1 is the range of experience from the clinical campaigns and not a binding specification, so both the decision rule below and the capability index described in Section 5.5 are judged against the upper bound alone. No viral clearance response arises at this step, so no step-level requirement has to be back-calculated from a cumulative requirement, and that case belongs to the viral steps (PCP-006, PCP-008 and PCP-009).

Two analyses will be run for each pair. The first analysis holds the other parameters at their set-points and evaluates the fitted response-surface model across the characterization range of the parameter of interest. The second analysis propagates the normal operating ranges of the other parameters by Monte-Carlo, drawing each of them inside its normal operating range and adding the model residual variation, on a grid of 81 points across the characterization range with 2,000 draws at each grid point. The random draw uses a fixed seed, so the reported ranges are reproducible from the same data.

The decision rule is the same in both analyses. The proven acceptable range is the contiguous interval of the parameter, containing the set-point, over which the attribute stays within its acceptance criterion. In the first analysis that means the predicted mean stays within the criterion, and in the second it means the ninety-five per cent predictive interval stays within the criterion, which is a robustness statement and is expected to give the narrower of the two ranges. Where the set-point condition itself fails the criterion, no proven acceptable range exists for that pair, and PCR-003 will report it as such.

Each governed attribute will be reported with a two-panel plot for its governing parameter, which is the parameter carrying the largest main effect on that attribute. The parameter is on the horizontal axis and the attribute on the vertical axis, the acceptance limit is drawn, the acceptable region is shaded green, and the set-point and the normal operating range are marked. The left panel shows the first analysis and the right panel shows the Monte-Carlo analysis with its predictive interval.

The relationship between these ranges and the design space will be stated explicitly in PCR-003. A proven acceptable range is a univariate statement about one parameter, and the second analysis extends it to co-variation of the others within their normal ranges. The design space is the multivariate region, and it is not the product of the individual proven acceptable ranges. Where a parameter's whole characterization range is proven acceptable under both analyses, the binding constraint is the worst-case corner of the design space and not that parameter's univariate range, and PCR-003 will say which constraint binds. Every range reported will be bounded to the characterization ranges in Table 6.1 and to the fitted model.

9 Data management and integrity

All study records will meet ALCOA+ expectations. Run set-points, control system trends and in-process measurements are captured in the bioreactor control system and its historian, and are transferred to the electronic laboratory notebook against the run identifier. Analytical raw data are captured in the chromatography data system and in the plate reader software for the immunoassays, with audit trails enabled and access controlled by role.

The design matrices, the executed conditions and the assembled response data set will be held as a version-controlled data set, and the statistical analysis will be executed from a version-controlled script under SOP-1002. That is what makes the reported effects, models and ranges reproducible from the raw data by a reviewer who was not present at the bench. Any change to the data set after analysis requires a documented reason, a re-run of the analysis and a second-person review.

Every record is reviewed by a second person before it is used in the report. Deviations, including any executed condition that departs from the design, are recorded in the quality system as they are detected and not retrospectively. Records are retained under the site retention policy.

10 Roles and responsibilities

The functions listed in Table 10.1 carry the study, and the approval record at the front of this document (Table 1) binds the same functions to the plan.

Table 10.1: Functions and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owns the plan, executes the scale-down runs, performs the analysis and writes PCR-003
Analytical Development	Maintains the validated methods, analyses the samples and reports assay performance
Manufacturing Science (receiving site)	Reviews the ranges and the control assumptions for feasibility at commercial scale
Statistics	Reviews the designs before execution and the fitted models before reporting
Quality Assurance	Approves this plan and PCR-003, and oversees deviation handling

The statistical review before execution is the last point at which the factor set, the ranges and the run count can still be changed. The review by manufacturing science at the receiving site checks the characterized ranges against what the commercial vessel can hold.

11 Deliverables and schedule

The deliverable of this plan is PCR-003, the process characterization report for the production bioreactor. It will contain the executed designs, the effect and coefficient tables, the model diagnostics, the design space, the proven acceptable ranges, the capability estimate, the classification of every parameter and any deviations recorded during execution. PCR-003 rolls up into the master report (PCMR-001) together with the reports of the other steps.

The work runs in five phases, and the order is fixed by dependency and not by convenience. Scale-down model qualification comes first and is approved before any characterization run. The screening campaign follows, and its analysis fixes the factor subset for the response-surface campaign (Section 6.3). The response-surface campaign and the univariate runs follow, and they may be interleaved as scale-down system capacity allows, because the univariate runs do not depend on the screening result. The capability and proven acceptable range analyses are performed on the fitted models once the response-surface data are complete, and the report is written last.

No calendar dates are committed in this plan. The schedule is held in the campaign plan (PCMP-001), and it depends on scale-down system availability and on the analytical throughput for the methods listed in Table 5.1.

12 Risks and assumptions

The study rests on assumptions that would change its conclusions if they proved false, and they are stated here so that PCR-003 can report against them.

The principal risk is that the scale-down model does not represent the commercial vessel over the whole characterized space. Qualification demonstrates equivalence at the target condition, and it does not demonstrate equivalence at the edges of the ranges, where mixing and gas transfer differ most between scales. This is managed in three ways, by matching the scale-dependent parameters on performance and not on geometry (Section 5.1), by reporting the pCO₂ and osmolality trajectories for every run, and by confirming the design space at commercial scale during Stage 2. The design space reported in PCR-003 will not be confirmed at scale at the edges of the ranges.

The second risk is analytical. The glycan responses come from a single separation (Section 5.3), so an assay shift moves 3 of the 5 responses together and can imitate a process effect. The centre-point replicates and the randomized run order are the controls for this, and any effect of the same size as the assay reproducibility will be reported as such.

The third risk is material. A change of basal medium or feed lot during the campaign introduces a variable that the design does not carry, and the effect of such a change on the glycan profile is not negligible. Lot usage will be recorded per run, so that a lot change can be tested as a term if the data suggest it.

Three assumptions are made explicit. Platform behaviour observed on the earlier products (X-Mab, Y-Mab and Z-Mab) is assumed to transfer to A-Mab, which is the basis for studying 5 parameters instead of the full list. The parameter list in Table 6.1 is assumed to be complete, which rests on the process description in PTP-001 and on the risk assessment. Three-factor interactions are assumed to be negligible, which is what allows the resolution V fraction described in Section 6.2. That assumption is carried on platform history, and the 3 residual degrees of freedom the screening model leaves are too few to test it. The control is the response-surface campaign, where a two-factor interaction that the screening analysis estimated and the response-surface model does not reproduce will be reported as evidence that the assumption was broken. A pair involving a factor that is not carried forward, such as osmolality under the subset anticipated in Section 6.3, cannot be checked in that way.

13 Approvals

This plan is effective when it has been approved by every function in the approval record at the front of the document (Table [1](#)). Execution may not begin before that approval. Any change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan under the site change control procedure.

14 Appendices A-B — Planned design matrices

The matrices below are the designs as planned. They carry no responses, because no data exist. The executed matrices with their measured responses are reported in the appendices of PCR-003. Factor letters follow Table 6.2, and the coded levels are minus one at the low edge of the characterization range, zero at the set-point and plus one at the high edge.

14.1 Appendix A — Planned screening design matrix

Table 14.1: Planned screening design matrix in coded units.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

14.2 Appendix B — Planned response-surface design matrix

Table 14.2: Planned face-centred central composite design matrix in coded units, drawn for the factor subset anticipated to be carried forward from screening.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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